

Retail Inventory Analytics Using Google Colab

PDF Report

Submitted By,

Name: Pavan Lagare

Course: Data Analytics

Project Title: Retail Inventory Analytics Using Google Colab

CONTENTS

Sr No	Title	Page No
1	Introduction	3
2	Project Objective	3
3	Dataset Overview	4
4	Data Cleaning	5
5	Exploratory Data Analysis	6
6	Business Insights	11
7	Recommendations	12
8	Conclusion	12

1. Introduction

Retail inventory management plays a crucial role in ensuring that products are available to customers while minimizing storage costs and stock shortages. Effective inventory analysis helps organizations monitor product performance, track stock levels, evaluate supplier contributions, and make informed business decisions.

In this project, Retail Inventory Analytics is performed using Python in Google Colab. Various data analysis and visualization techniques are applied to understand inventory trends, identify low-stock products, analyze sales performance, and generate actionable business insights. The project demonstrates how data analytics can support efficient inventory management and improve overall business operations.

2. Project Objective

The primary objective of this project is to analyze retail inventory data using Python and Google Colab to gain meaningful insights into inventory performance.

The specific objectives are:

- To identify product categories that generate the highest sales.
- To detect products that are approaching reorder levels.
- To analyze the relationship between stock availability and product sales.
- To evaluate supplier contributions to inventory distribution.
- To create visual dashboards and charts for better decision-making.
- To provide business recommendations for improving inventory management efficiency.

3. Dataset Overview

The Retail Inventory dataset contains inventory information for products belonging to multiple categories and suppliers. The dataset is used to analyze sales performance, stock availability, supplier contribution, and inventory trends.

Dataset Summary

- Total Records: 1000
- Total Attributes: 11
- Missing Values: 0
- Duplicate Records: 0

Dataset Attributes

1. Product ID – Unique identifier for each product.
2. Product Name – Name of the product.
3. Category – Product category.
4. Supplier Name – Supplier providing the product.
5. Unit Cost – Cost price of the product.
6. Retail Price – Selling price of the product.
7. Opening Stock – Initial stock available.
8. Units Sold – Number of units sold.
9. Closing Stock – Remaining stock after sales.
10. Reorder Point – Stock level at which replenishment is required.
11. Stock Status – Current inventory status (In Stock / Low Stock).

The dataset contains both numerical and categorical variables, making it suitable for inventory analysis, sales trend identification, stock monitoring, and business decision-making.

4. Data Cleaning

Data cleaning was performed to ensure the accuracy and reliability of the dataset before conducting analysis.

Duplicate Record Check

The dataset was checked for duplicate records using Python. No duplicate records were found, indicating that each inventory record is unique.

Result: 0 Duplicate Records Found

Missing Value Check

All columns were examined for missing values. The analysis confirmed that there are no null or missing values present in the dataset.

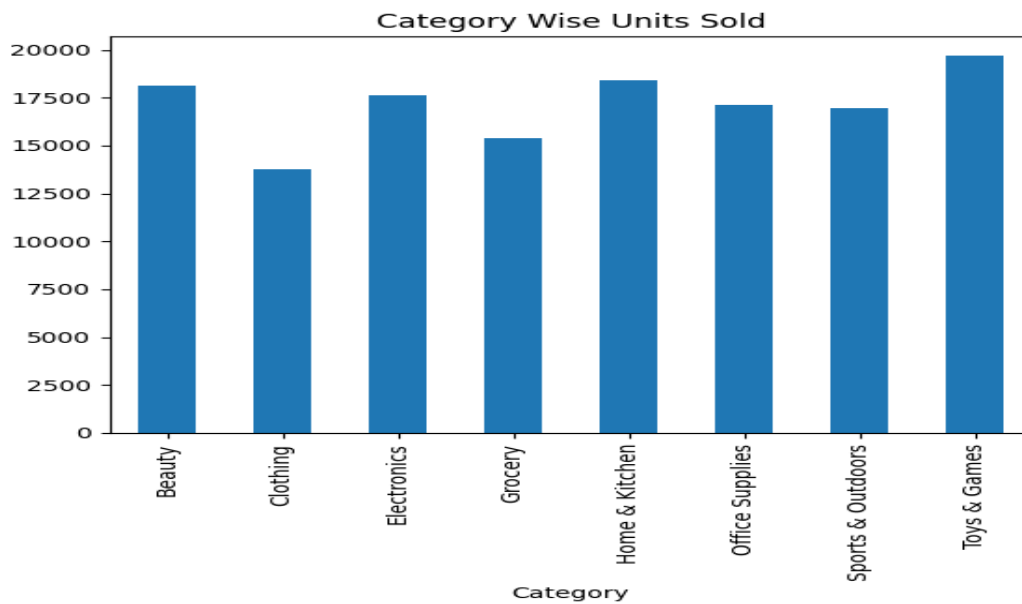
Result: 0 Missing Values Found

Stock Data Validation

Stock-related fields such as Opening Stock, Units Sold, Closing Stock, and Reorder Point were validated using descriptive statistics. All values were found to be logical and within acceptable ranges.

5. Exploratory Data Analysis and Visualizations

5.1 Category Wise Sales



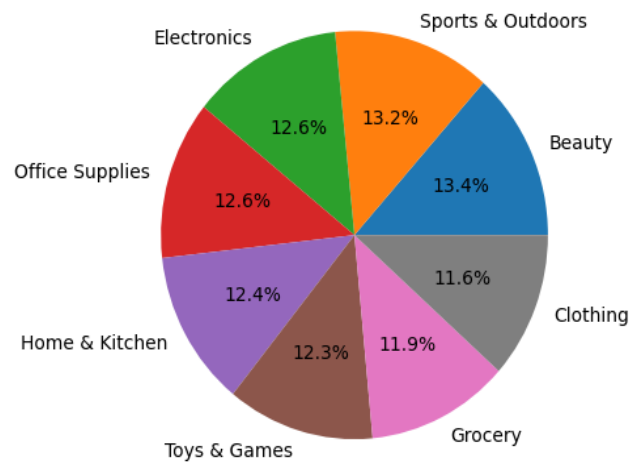
Observation

- Toys & Games generated the highest sales among all categories.
- Clothing generated the lowest sales among all categories.

Conclusion

The Toys & Games category is the fastest-moving category and contributes significantly to overall sales.

5.2 Category Distribution



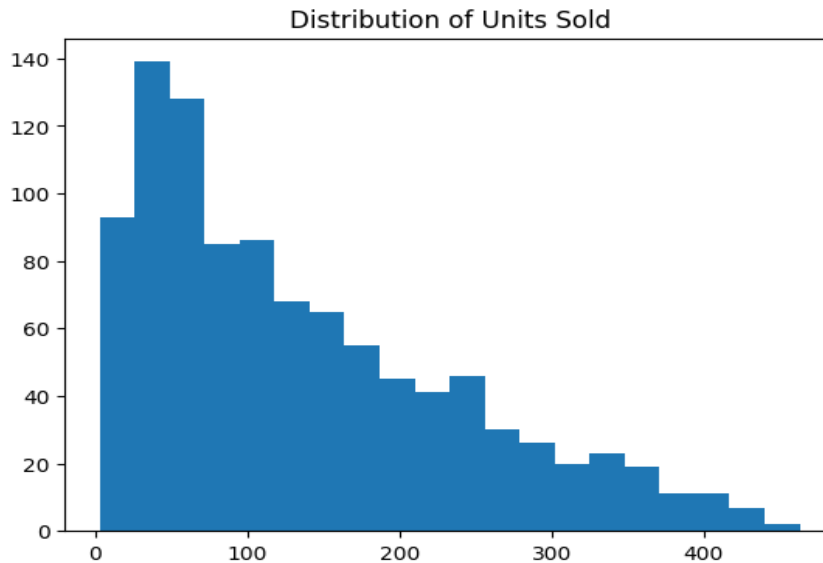
Observation

- Product distribution across categories is relatively balanced.
- Beauty category has the highest share of products.

Conclusion

The company maintains a diversified inventory across all product categories.

5.3 Distribution of Units Sold



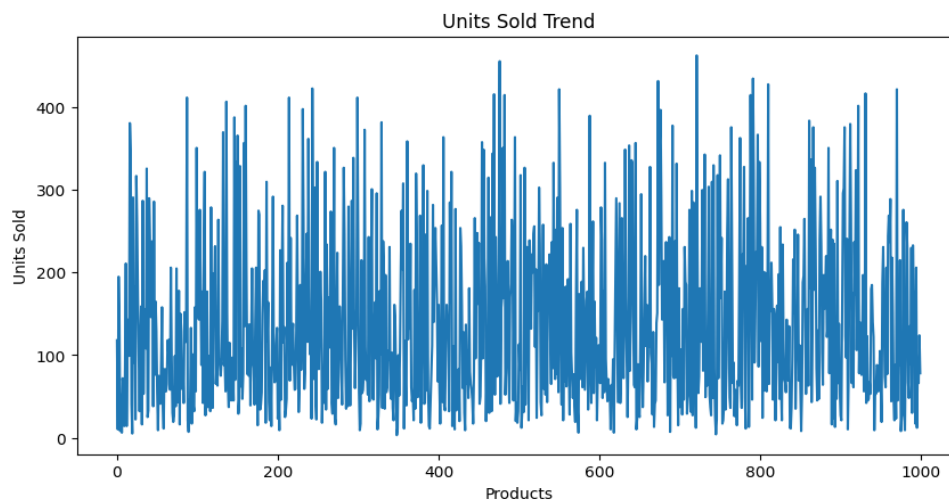
Observation

- Most products recorded sales between 0 and 150 units.
- Only a small number of products achieved very high sales volumes.

Conclusion

The majority of products experience moderate sales, while a few products contribute significantly to total sales.

5.4 Top Selling Products



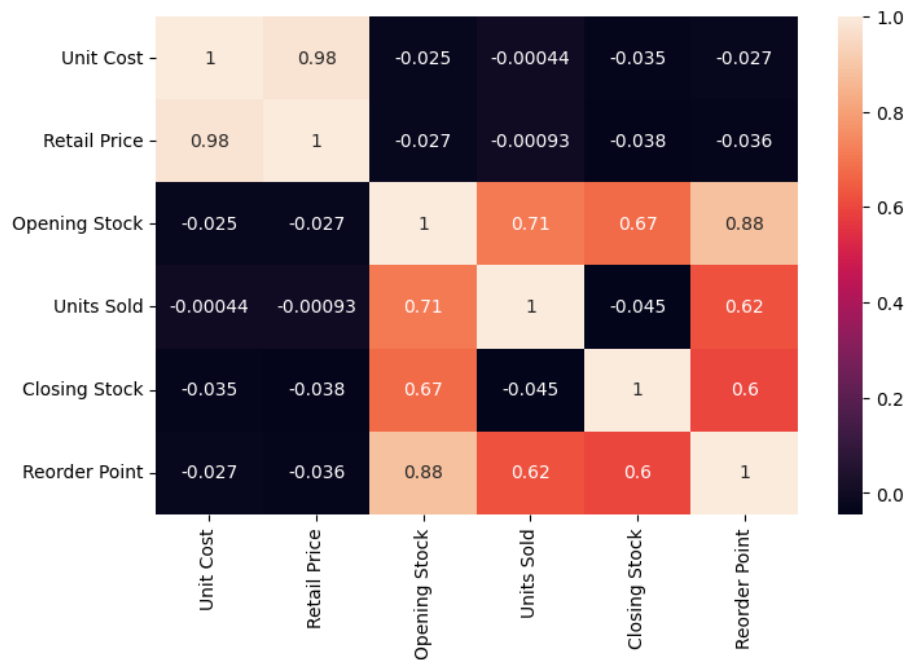
Observation

- Several products have significantly higher sales than others.
- Top-selling products contribute a major share of overall revenue.

Conclusion

High-demand products should be closely monitored to avoid stock shortages and maintain customer satisfaction.

5.5 Correlation Heatmap



Key Findings

Variables	Correlation
Unit Cost & Retail Price	0.98
Opening Stock & Reorder Point	0.88
Opening Stock & Units Sold	0.71
Units Sold & Reorder Point	0.62

Observation

- Unit Cost and Retail Price show a very strong positive correlation.
- Opening Stock is positively related to Units Sold and Reorder Point.

Conclusion

Products with higher stock availability generally achieve higher sales, highlighting the importance of effective inventory planning.

6. Business Insights

1. Products Requiring Immediate Replenishment

Products with low closing stock and low-stock status should be replenished immediately to avoid inventory shortages.

2. Fast-Moving Product Categories

The Toys & Games category recorded the highest sales and requires frequent inventory monitoring.

3. Slow-Moving Inventory

Clothing and other lower-selling categories may need promotional strategies to improve sales performance.

4. High-Profit Opportunities

Products with a higher difference between retail price and unit cost provide better profit opportunities for the business.

5. Supplier Dependency Risk

Inventory is heavily dependent on a few suppliers, which may increase supply chain risks if disruptions occur.

7. Recommendations

- Maintain higher inventory levels for fast-moving categories such as Toys & Games and Home & Kitchen.
- Regularly monitor products approaching reorder levels to avoid stock shortages.
- Implement automated reorder alerts for low-stock products.
- Review slow-moving inventory and introduce promotional strategies to improve sales.
- Diversify suppliers to reduce dependency risks and ensure a stable supply chain.

8. Conclusion

The Retail Inventory Analytics project successfully analyzed retail inventory data using Python and Google Colab. The dataset was found to be clean, with no missing values or duplicate records. Exploratory Data Analysis and visualizations helped identify high-selling product categories, stock distribution patterns, supplier contributions, and products approaching reorder levels.

The analysis revealed that Toys & Games was the highest-performing category, while several products required closer inventory monitoring due to low stock levels. Correlation analysis also showed a positive relationship between stock availability and sales performance.

Overall, the project demonstrates how data analytics can support effective inventory management, improve stock planning, reduce inventory risks, and assist management in making data-driven business decisions.